

In the `__init__()` method ❶, we set up a number of instance variables to use in the methods of the class. That lets us keep the code of the methods very simple. Since the `__init__()` method saved the rectangle of the `Square`, the `clickedInside()` method ❷ just checks if the location of a mouse click was inside that rectangle, returning `True` OR `False`.

The `getType()` method ❸ simply returns the information that the item clicked is a square. The `getArea()` method ❹ multiplies the width by the height and returns the resulting area. The `draw()` method ❺ uses pygame's `draw.rect()` to draw the shape in the randomly chosen color.

The Circle and Triangle Shape Classes

Next, let's take a look at the code of the `Circle` and the `Triangle` classes. The important thing to notice is that these classes have methods with the same names as the `Square` class, but the code in these methods (especially `clickedInside()` and `getArea()`) is very different. [Listing 9-3](#) shows the `Circle` class. [Listing 9-4](#) shows the `Triangle` class that creates randomly sized right triangles whose edges are parallel to the x- and y-axes, with the right angle in the upper-left corner.

File: Shapes/Circle.py

```
# Circle class

import pygame
import random
import math

# Set up the colors
RED = (255, 0, 0)
GREEN = (0, 255, 0)
BLUE = (0, 0, 255)

class Circle():

    def __init__(self, window, maxWidth, maxHeight):
        self.window = window

        self.color = random.choice((RED, GREEN, BLUE))
```